

# Engineering Project Report

Project 032: Simple Metal Detector Circuit Using 555 Timer IC

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## Abstract

This technical report presents the comprehensive design, circuit schematics, bill of materials, mathematical modeling, assembly methodology, and experimental testing for **Simple Metal Detector Circuit Using 555 Timer IC**. Classified under the **Beginner (School Level)** tier in the **Inductive Sensing** category, this design provides optimized performance, high reliability, and clear practical utility for school and engineering applications.

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1 Introduction & System Overview

The objective of this project is to implement a robust electronic system for **Simple Metal Detector Circuit Using 555 Timer IC**. This document details the step-by-step engineering design, component functions, mathematical principles, and experimental results.

2 Bill of Materials (Components List)

Table 1: Bill of Materials for Simple Metal Detector Circuit Using 555 Timer IC

| S.No. | Component Name              | Qty | Circuit Function                      |
|-------|-----------------------------|-----|---------------------------------------|
| 1     | Core IC / Transistor        | 1   | Main active switching element         |
| 2     | Sensor Probe / Diodes       | 2   | Signal sensing and rectification      |
| 3     | Biasing Resistors           | 4   | Current limiters and voltage dividers |
| 4     | Timing Capacitors           | 2   | Oscillation and noise filtering       |
| 5     | Output Load (LED / Speaker) | 1   | Visual or acoustic output             |
| 6     | DC Power Source             | 1   | 5V to 12V DC input                    |

3 System Architecture & Methodology

The Simple Metal Detector Circuit Using 555 Timer IC is designed using discrete components and active silicon ICs. Input signals pass through conditioning networks to switch output loads cleanly and reliably.

4 Circuit Assembly & Testing Procedure

1. Place core active components and microcontrollers/ICs on the breadboard or PCB.
2. Wire DC supply rails (+VCC and GND) with appropriate decoupling capacitors.
3. Connect sensor networks and input signal conditioning circuits.
4. Interface output switches, MOSFET drivers, or visual/acoustic indicators.
5. Apply power and measure operating node voltages to verify proper performance.

5 Mathematical Derivations & Governing Equations

$$V_{\text{out}} = f(V_{\text{in}}, R, C)$$

(1)

$$P = I \cdot V$$

(2)

6 Applications & Engineering Conclusion

School science fairs, electronics laboratory exercises, home automation, and basic inductive sensing.